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Association analysis of sugar yield-related traits in sorghum [*Sorghum bicolor* (L.)]

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Abstract Association mapping is widely used for detecting QTLs in higher plants. In the present study a synthetic sorghum population containing 119 representative samples, including 43 sweet and 76 grain sorghum accessions originating mainly from China, USA and India, were genotyped using 51 simple-sequence repeat (SSR) markers. Linkage disequilibrium (LD) of pair-wise loci and population structure were analyzed, followed by association analysis of SSR loci and 3 sugar yield related traits using the TASSEL general linear model program. Results showed that: (i) different degrees of LD occurred among syntenic markers and also among nonsyntenic markers, indicating historical recombination among sorghum linkage groups; (ii) significant LD extended up to 7.31 cM; (iii) the collection of accessions was composed of three subgroups; (iv) four marker loci were associated with stalk sugar concentration, fresh stalk weight and stalk juice weight measured in different growing environments and could be used, therefore, in future marker assisted breeding programs. Several loci were also associated with two or more traits simultaneously, which might be due to tight linkage between different genes affecting these traits and/or pleiotropy. In addition, some associated markers were located close to QTLs previously mapped in family-based linkage mapping analyses.

Keywords *Sorghum bicolor* · SSR · Population structure · Association mapping

Abbreviations

LD	Linkage disequilibrium
SSR	Simple sequence repeat
QTL	Quantitative trait loci
FSW	Fresh stalk weight
SJW	Stalk juice weight

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Introduction

Bioenergy, derived from fixed solar energy through plant photosynthesis, is an important renewable resource for producing clean fuel, for reducing environmental pollution and also dependency on non-renewable resources. Currently, energy crops are mainly represented by perennial grasses such as switchgrass (*Panicum virgatum* L.), energy cane (*Saccharum* spp.), sweet and forage sorghum (*Sorghum bicolor*), miscanthus (*Miscanthus* spp.), as well as some short-rotation forest resources, e.g. *willow*—*Salix* spp. and *poplar*—*Populus* spp. (Jessup 2009; McCutchen et al. 2008). Of these, sweet sorghum has emerged as one of the most promising energy crops. The soluble sugar (sucrose, glucose and fructose) content accumulated in the stalks of this plant can reach up to 19 % of stem fresh weight (Wang and Liu 2009) and can be directly fermented into ethanol or other forms of biological fuel with conversion efficiencies of more than 90 % (Wu et al. 2010). Up to 13.2 t/ha of total sugars, equivalent to 7,682 l of ethanol per hectare can be produced by sweet sorghum under favorable conditions (Jackson et al. 2008). Moreover, its crushed stems (bagasse) can be processed as lignocellulosic biomass, which can be converted to ethanol or used in other traditional ways. Sorghum is tolerant to heat, salinity and drought, producing high biomass under such conditions. It has a short growing season, usually 3–5 months, and therefore can be harvested 2–3 times annually. In addition, sorghum is propagated by seed, and contains varieties and hybrids adapted to a wide range of different environmental conditions. In combination, these features make sorghum one of the most promising of all bioenergy crops.

Sweet and grain sorghums are very similar for most characters; however, sweet sorghums produce more biomass, exhibit more rapid growth and wider adaptation (Corn 2009), and have juicier stalks with higher levels of sugar in the juice. The sugar productivity of sweet sorghum is mainly determined by sugar concentration and stalk juice yield. Previous studies have indicated that sugar concentration is affected by many genes (i.e. is subject to polygenic control), and by gene–environment interactions. Some quantitative trait loci (QTLs) affecting sugar concentration have been isolated in different sorghum mapping populations and found to be located on chromosomes 1, 2, 3, 4, 5 and 7 (Bian et al. 2006; Brown et al. 2006; Felderhoff et al. 2012; Guan et al. 2011; Lekgari 2010; Murray et al. 2008a, b, 2009; Natoli et al. 2002; Ritter et al. 2007; Shiringani et al. 2010). For example, one QTL on chromosome 3 was identified in several different mapping populations, and shown to explain between 18 and 25 % of total phenotypic variation in sugar concentration (Felderhoff et al. 2012; Murray et al. 2008b; Natoli et al. 2002). However, the presence of this locus was not confirmed in a subsequent association mapping study that involved analysis of 125 sweet sorghum varieties including a sweet sorghum variety, Rio, which was one of the parental lines used in previous family-based QTL mapping studies.

A previous mapping study also detected a QTL peak on chromosome 1, which explained up to 9 % of variation for brix and sugar concentration; however, this QTL fell slightly below the stringent threshold for significance (Murray et al. 2008b). In a follow-up study the region was confirmed to be significantly associated with brix, and shown to be located 12 kb from a sorghum homolog of glucose-6-phosphate isomerase (EC 5.3.1.9), suggested to be a candidate gene (Murray et al. 2009). Lekgari (2010) failed to detect a QTL on chromosome 3 in his mapping study, but found one on chromosome 1 that explained 33.9 % of trait variation.

Another factor determining sugar production is stalk juice yield with juicy versus dry stem considered to be controlled by a major gene (Bennetzen et al. 2001). However, some recent studies have shown that more than one QTL determines stalk juice yield (Guan et al. 2011; Lekgari 2010). Thus, in a recombinant inbred line (RIL) mapping population derived from a cross between a grain [09178(1)] and a sweet sorghum (N99) line, two QTLs located on chromosome 1 and 6, respectively, were isolated and shown to explain 77.0 and 7.7 % of trait variation, respectively (Lekgari 2010). However, the effects of these QTLs were not detected in all environments indicating that gene-environment interactions also affect performance. Although some QTLs linked with brix and stem juice yield have been identified, only a few sweet sorghum varieties have been screened using family-based mapping methods. Thus, there is a need to screen breeding material more extensively and effectively to evaluate the number and individual effects of genes controlling these important traits.

Recently, association or linkage disequilibrium mapping has become a popular method for dissecting the genetic basis of complex traits in plants. Compared with QTL mapping that may involve difficulties in mapping population construction in some species, the presence of only one meiotic generation, and genetic diversity derived from only two parents, association mapping uses naturally occurring variation in diverse germplasms and does not suffer, therefore, from a low level of variation that characterizes some family-based QTL mapping populations. In addition, naturally occurring recombination will have occurred over many generations and consequently linkage blocks will be substantially smaller in an association mapping population thus enabling more fine-scale mapping of genes controlling traits (Hall et al. 2010). Early use of association mapping in plants involved a candidate gene approach in maize (Thornsberry et al. 2001) and was subsequently applied successfully to wheat (Breseghello and Sorrells 2006), rice (Agrama et al. 2007), barley (Kraakman et al. 2004) and sugarcane (Wei et al. 2006), as well as sorghum (Casa et al. 2008; Morris et al. 2013; Shehzad et al. 2009) and sweet sorghum (Murray et al. 2009). In the present study, association mapping was conducted on a large panel of sorghum germplasms that partly included landraces and elite varieties from North China. This approach was used to investigate QTLs associated with bioenergy traits.

Materials and methods

Plant materials and method of cultivation

A panel of 119 sorghum accessions was investigated which included 55 landraces from North China and 64 elite varieties originating from the USA (45), India (10), Japan (5), Mexico (2) and Romania (2). Forty-three of these accessions represented sweet sorghum and 76 represented grain sorghum germplasm (Table 1). Accessions were planted at the Shijiazhuang and Haixing Experimental Stations, China, in 2010 and 2011. Soil condition were normal at Shijiazhuang, but were typically saline and alkaline at Haixing. All accessions were planted in 2 replicated, randomized complete blocks, with 6 plants of each accession per replication. Planting density was 25 × 10 cm. Climatic conditions in 2011 were such that some plants did not reach full maturity at Haixing, and consequently results for these plants were not included in statistical analysis.

Molecular marker analysis

DNA was extracted from young leaves using the CTAB method of Doyle (Doyle and Doyle 1987). A total of 51 SSR markers known to be evenly distributed across all 10 sorghum chromosomes were used to genotype plants (Table 2). All primers were selected according to currently available genetic maps (http://sorgblast3.tamu.edu/linkage_groups.htm). The PCR reaction system was composed of 50 ng genomic DNA, 100 ng primer pair, 125 μM dNTP, 50 mM KCl and 10 mM Tris–HCl, 2 mM $MgCl_2$, and 1 unit Taq polymerase. The amplification procedure consisted of one cycle at 94 °C for 3 min, followed by 35 cycles of 1 min at 94 °C, 1 min at 55 to 58 °C depending on the primer pair, 1 min at 72 °C, and a final extension step at 72 °C for 8 min. The PCR products were separated on a 5 % polyacrylamide gel followed by silver staining.

Phenotypic evaluation

Plants were harvested manually at both locations and in both years at the dough stage (30 to 40 days after flowering) by cutting plants near to the soil surface. At each experimental site, two plants of each accession were collected for phenotypic evaluation. Plants from the borders of a plot were avoided and thus plants for analysis were sampled randomly from the remainder of the population. Each sampled plant was measured for (i) fresh stalk weight after the panicle was cut at the flag leaf and all leaves removed, (ii) stalk juice weight, estimated after fresh stripped stalks were milled twice by a roller miller, and (iii) brix reading obtained by measuring 2–3 droplets of juice using a hand-held refractometer PAL-1 (ATAGO Co., Itabashi-ku, Japan). The average values per trait of the two plants per accession were used for analysis.

Data analysis

The number of alleles values of gene diversity and polymorphism information content (PIC), and the genetic distance between individuals (Nei et al. 1983) at each of the 51 SSR marker loci were calculated using software PowerMarker version 3.25 (Liu and Muse 2005). PIC was calculated using the formula $PIC = 1 - \Sigma\ (Pi)^2$, where Pi is the proportion of the population carrying the ith allele. An unrooted phylogenetic tree was constructed from genetic distances between individuals using software FigTree 1.4 (http://tree.bio.ed.ac.uk/software/figtree/).

Population structure among accessions was assessed using a model-based (Bayesian) clustering algorithm implemented in structure 2.3.4 (Pritchard et al. 2000). Structure was analyzed by examining a range of possible Ks (an assumed fixed number of subpopulations) from 1 to 10 in the entire population. Five independent analyses were conducted for each k value with the burn-in period equal to 50,000 iterations and a run of 100,000 replications of markov

Table 1 List of germplasm accessions used

Entry	Name	Designation	Character	Subgroup	Origin
12	Lutian		Sweet	1	Beijing
18	Tianx190		Sweet	1	Beijing
104	Tianx89		Sweet	1	Beijing
112	Tianx3		Sweet	1	Beijing
2	1907		Sweet	1	Huanghua
3	1910		Sweet	1	Huanghua
4	DH		Sweet	1	Huanghua
14	Huangh1		Sweet	1	Huanghua
87	Huangh2		Sweet	1	Huanghua
101	K1453		Sweet	1	Huanghua
102	1908		Sweet	1	Huanghua
110	K1456		Sweet	1	Huanghua
32	J192	JT2R3P1	Grain	1	India
34	J184	J14636	Grain	1	India
35	J183	J14625	Grain	1	India
45	JG51	MIV22787	Grain	1	Mexico
47	JG30	TX2761	Grain	1	Mexico
11	L167		Sweet	1	Shanxi
15	Gaot2		Sweet	1	Shanxi
85	Jinl5		Grain	1	Shanxi
90	K773		Grain	1	Shanxi
100	Sxzs1		Grain	1	Shanxi
111	Ltian3		Sweet	1	Shanxi
17	Jingxt		Sweet	1	Shijiazhuang
66	R172		Grain	1	Shijiazhuang
109	Keter		Sweet	1	Shijiazhuang
7	Ximeng		Sweet	1	USA
8	Bailty	NSL187557	Sweet	1	USA
10	T132		Grain	1	USA
16	US-WD1		Sweet	1	USA
23	JG12	ICCH23	Grain	1	USA
28	JG13	ICSV393	Grain	1	USA
29	JG14	IS-18704	Grain	1	USA
30	Yitian		Sweet	1	USA
53	JG29	ES230	Grain	1	USA
65	Kaili		Sweet	1	USA
68	Sart	NSL91616	Sweet	1	USA
78	J201	J14725	Grain	1	USA
80	J197	J14694	Grain	1	USA
89	JG47	P932127	Grain	1	USA
92	Roma		Sweet	1	USA
103	Wray		Sweet	1	USA
105	Cowley		Sweet	1	USA
106	Rio		Sweet	1	USA
115	J176-1		Sweet	1	USA
117	113-39		Grain	1	USA

Table 1 continued

Entry	Name	Designation	Character	Subgroup	Origin
118	J172	VRGA	Grain	1	USA
6	Tashib		Sweet	2	N/A
21	Qiubai		Sweet	2	N/A
48	Jiliang2		Grain	2	Baoding
13	Bj339		Sweet	2	Beijing
20	T129		Grain	2	Beijing
91	Lun1		Sweet	2	Beijing
52	KB14		Grain	2	Chengde
61	R431		Grain	2	Chengde
62	R129		Grain	2	Chengde
64	Tiangl		Seet	2	Chengde
67	R138		Grain	2	Chengde
108	KB12		Grain	2	Chengde
113	B25		Grain	2	Chengde
9	JG36	IS634	Grain	2	India
24	JG11	ICSB43	Grain	2	India
25	JG21	IS5483	Grain	2	India
26	JG17	F416009	Grain	2	India
27	JG15	IS18704	Grain	2	India
44	JG22	IS5613	Grain	2	India
49	JG35	ISCB88016	Grain	2	India
50	JG38	DWARD	Grain	2	Japan
96	JG42		Grain	2	Japan
99	JG41		Grain	2	Japan
1	Luotz		Sweet	2	Romania
107	Luotz13t		Sweet	2	Romania
58	Chun60		Sweet	2	Shanxi
19	USWD2		Sweet	2	USA
22	JG50	MN567	Grain	2	USA
33	J186	J14638	Grain	2	USA
46	JG32	MIV4323	Grain	2	USA
51	JG24	IVK133	Grain	2	USA
54	JG33		Grain	2	USA
55	JG31	MIV3091	Grain	2	USA
74	J208	J14796	Grain	2	USA
76	J207	J14775	Grain	2	USA
77	J190	J14657	Grain	2	USA
81	J196	J14693	Grain	2	USA
82	J198	J14695	Grain	2	USA
83	J203	J14757	Grain	2	USA
114	J148		Grain	2	USA
115	J161	JP313-2	Grain	2	USA
119	J169	JP412-2	Grain	2	USA
70	Tianx11		Sweet	3	Beijing
71	Tianx60		Sweet	3	Beijing

Table 1 continued

Entry	Name	Designation	Character	Subgroup	Origin
31	R167		Grain	3	Chengde
37	R498		Grain	3	Chengde
38	R419		Grain	3	Chengde
39	30111		Grain	3	Chengde
41	R44		Grain	3	Chengde
42	R505		Grain	3	Chengde
43	R305		Grain	3	Chengde
57	R526		Grain	3	Chengde
59	R1311		Grain	3	Chengde
63	K1455		Sweet	3	Chengde
69	B5		Grain	3	Chengde
72	R156		Grain	3	Chengde
73	R380		Grain	3	Chengde
86	R271		Grain	3	Chengde
94	R385		Grain	3	Chengde
95	R425		Grain	3	Chengde
97	JG39	MARTIN	Grain	3	Japan
98	JG40	WINNER	Grain	3	Japan
36	Xingjt		Sweet	3	Shijiazhuang
5	Yxgl1		Grain	3	Shijiazhuang
40	JG26	Merasi	Sweet	3	USA
56	JG25	Honey-4	Sweet	3	USA
60	J213	JTRIPH	Grain	3	USA
75	J209	JTIRIPI	Grain	3	USA
79	J200	JI4720	Grain	3	USA
84	J205	J14771	Grain	3	USA
88	J193	J14663	Grain	3	USA
93	JG44	IS1291	Grain	3	USA

chain monte carlo (MCMC) after burn in. The most likely k value was determined using the method of Evanno et al. (2005). The average matrix of ancestry membership proportions of the 5 runs (Q matrix) were computed using CLUMPP version 1.1 (Jakobsson and Rosenberg 2007). All accessions were assigned to a subpopulation according to the largest probability estimated by the program.

Linkage disequilibrium (LD) and genotype-phenotype associations

Linkage disequilibrium was calculated for each pair of SSR loci using TASSEL 2.1 software (Bradbury et al. 2007). Significance (*p* values) of D′ for each SSR pair was determined by 10,000 permutations. LD was estimated separately for each chromosome, then the genome-wide LD decay was estimated by plotting the pair-wise r^2 values against the genetic distance (cM) from all 10 chromosomes. Curve of LD decay was fitted using non-linear regression (NLR) and a trend line was drawn using SPSS18. D′ ranges from 0 to 1 with 0.5 as a threshold of decay. The significant level was set at $p < 0.05$.

Genotype-phenotype associations were evaluated using the general linear model (GLM) of TASSEL, with SSR markers as fixed effects and elements of the Q matrix as covariates. *p* values were generated for each test using 10,000 permutations of genotypes with respect to phenotypic trait values.

Table 2 Information on the SSR markers used in study

Marker	LG	Position (cM)	Repeat motifs	Size (bp)	Allele no	Heterozygosity	PIC
xtxp482	1	25.2	$(GA)_{16}$	234	5	0.0536	0.6538
xtxp302	1	31.7	$(TGT)_8$	180	5	0.0137	0.5957
xtxp329	1	102.1	$(ATC)_8 + (CTT)_{22}$	162	17	0.0690	0.8206
xtxp149	1	106.2	$(CT)_{10}$	169	3	0.0000	0.3562
xtxp37	1	133.2	$(TC)_{23}$	189	10	0.0000	0.8170
xtxp335	1	137.1	$(GT)_{12}$	174	4	0.0000	0.6869
xtxp284	1	191.1	$(AAG)_{19}$	219	7	0.0000	0.7700
xtxp340	1	219.6	$(TAC)_{15}$	198	5	0.0087	0.7447
xtxp197	2	10	$(AC)_{10}$	154	3	0.0172	0.5372
xtxp201	2	74	$(GA)_{36}$	222	8	0.0000	0.7221
xtxp56	2	118.5	$(GA)_{39}$	349	6	0.0250	0.7566
xtxp286	2	128	$(GCA)_4ACA(GCA)_5A(CAA)_5 + (AAC)_9$	197	4	0.0000	0.5613
xtxp228	3	13	$(TC)_{12}$	246	3	0.0420	0.5532
xtxp205	3	85	$(AG)_{12}$	211	3	0.0000	0.4757
xtxp59	3	120	$(GGA)_5$	207	3	0.0000	0.1604
Cba	3	128	$(TA)_{18}$	217	3	0.1348	0.3722
xtxp114	3	138	$(AGG)_8$	234	3	0.0268	0.4877
xtxp69	3	187	$(TC)_{12}$	188	7	0.0330	0.6337
xtxp504	4	5	$(CAG)_7$	170	5	0.0336	0.3954
xtxp60	4	136	$(GT)_4GC(GT)_5$	224	3	0.0000	0.4432
xtxp212	4	141	$(GT)_{10}$	150	4	0.0336	0.4871
xtxp51	4	142.4	$(TG)_{11}$	234	4	0.0000	0.5279
xtxp65	5	11.2	$(ACC)_4 + (CCA)_3CG(CT)_8$	128	8	0.0603	0.7910
xtxp303	5	37	$(GT)_{13}$	160	7	0.0588	0.7651
xtxp123	5	115.8	$(AT)_9$	279	4	0.0435	0.5905
Kaf2	5	139.1	$(CAA)_9$	279	5	0.0504	0.5852
xtxp6	6	0	$(CT)_{33}$	134	5	0.0175	0.6449
xtxp145	6	53	$(AG)_{22}$	238	8	0.0342	0.7247
xtxp265	6	57	$(GAA)_{19}$	209	9	0.0326	0.8218
xtxp274	6	59	$(TTC)_{19}$	331	3	0.0291	0.4956
xtxp97	6	62	$(CA)_8 + (GCC)_6$	130	3	0.0000	0.4109
xtxp176	6	80	$(AG)_4AAC(GA)_4$	161	6	0.0513	0.3551
xtxp17	6	98.8	$(TC)_{16} + (AG)_{12}$	164	6	0.0000	0.5995
xtxp312	7	45.9	$(CAA)_{26}$	192	10	0.0194	0.7829
xtxp278	7	79.1	$(TTG)_{12}$	249	2	0.0169	0.3302
xtxp295	7	141	$(TC)_{19}$	165	6	0.0000	0.7147
xtxp168	7	160	$(AC)_{10}$	178	6	0.0261	0.5550
xtxp273	8	0	$(TTG)_{20}$	223	6	0.0171	0.5608
xtxp47	8	38.7	$(GT)_8(GC)_5 + (GT)_6$	264	3	0.0098	0.4442
xtxp354	8	97	$(GA)_{21} + (AAG)_3$	157	13	0.0259	0.8006
xtxp105	8	126.1	$(TG)_5 + (CT)_6GTCT(GT)_7$	291	2	0.0086	0.3727
xtxp358	9	20	$(ATT)_{22}$	267	5	0.0000	0.6522
xtxp287	9	43.6	$(AAC)_{21}$	367	5	0.0172	0.4772
xtxp258	9	45	$(AAC)_{19}$	222	10	0.0000	0.4780
xtxp230	9	53	$(GA)_{28}$	191	11	0.0000	0.8685

Table 2 continued

Marker	LG	Position (cM)	Repeat motifs	Size (bp)	Allele no	Heterozygosity	PIC
xtxp10	9	83	$(CT)_{14}$	145	5	0.0265	0.6404
Undhsbm105	10	22	$(GA)_{13}$	186	7	0.0504	0.6997
xtxp217	10	69	$(GA)_{23}$	175	4	0.2018	0.5333
PepC	10	75	$(AT)_{10}$	247	8	0.0294	0.7572
Undhsbm313	10	108	$(TAAA)_{13}$	170	8	0.0504	0.7328
xtxp141	10	129	$(GA)_{23}$	163	11	0.0085	0.8253

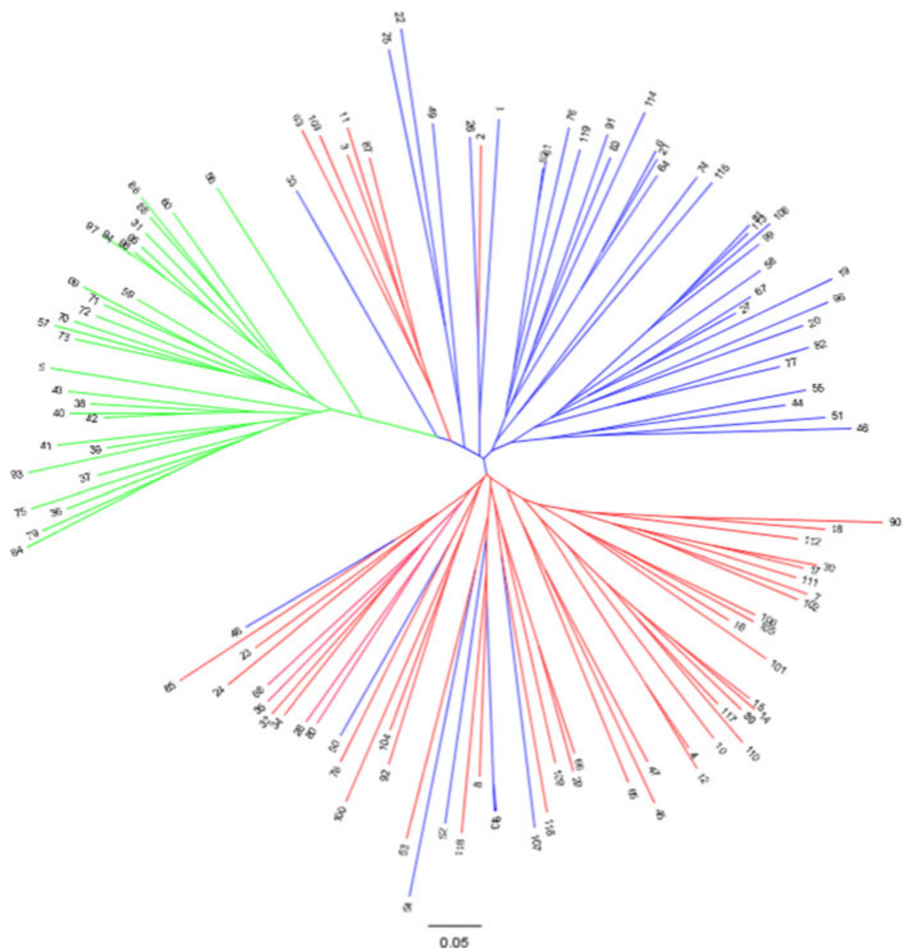

Fig. 1 Dendrogram showing relationships among 119 sorghum accessions revealed by cluster analysis based on neighbor-joining (NJ) of genetic distance. The colors of the unrooted tree show the different subgroups pop1 (*red*), pop2 (*blue*) and pop3 (*green*). Numbers represent the accessions according to Table 1

Results

Population structure

Fifty-one SSR markers were employed to analyze genetic diversity among 119 sorghum accessions. A total of 301 alleles were identified ranging from 2 (xtxp105) to 17 (xtxp329) per locus with an average of 5.9 alleles per locus across all loci. The polymorphism information content (PIC) value ranged from 0.1604 (xtxp59) to 0.8685 (xtxp230) with an average of 0.5994 across loci. Average heterozygosity was 2.7 % and ranged from 0 to 20.18 % (xtxp217).

The analysis of genetic structure among accessions using structure and evaluated using the ΔK test of Evanno et al. (2005) indicated that the collection of accessions was composed of three sub-populations (K = 3) with 92.4 % of the accessions strongly placed into these three groups. The three groups (pop1, pop2 and pop3) contained 47, 42 and 30 accessions, respectively (Table 1). With regard to Chinese accessions, all landraces from Huanghua county were assigned to pop1. Huanghua, near Haixing experimental station, is located at costal area where the soil is typical saline and alkali, and temperature is low throughout the sorghum growing season. The landraces originated from this region have short growing period, strong salt tolerance and lodge resistance, along with most varieties from Shanxi province, while most landraces from the Chengde area in north Hebei province were assigned to pop3 with common features of short life span, rapid growth rate, low height and small spike. Others with much higher plant height and longer growing period were assigned to pop2. However, all the varieties from this area showed strong cold and drought tolerance.

An unrooted tree constructed from pair-wise genetic distances among the 119 sorghum accessions (Fig. 1) showed that all accessions were divided into three major groups in a very similar way to that revealed by the structure analysis.

Linkage disequilibrium

The LD among the 51 SSR markers distributed across the 10 sorghum chromosomes is illustrated in Fig. 2. Different degrees of LD were detected (Fig. 2, upper triangle) not only among syntenic markers, but also among nonsyntenic ones, across the 1275 pair-wise comparisons among SSR loci. This suggests that historical recombination has occurred among linkage groups. However, only a few high level linkage disequilibrium values ($D' > 0.5$), i.e. 9.4 % (120 out of 1,275), were detected across the entire genome. Furthermore, the percentage of loci showing significant LD ($p < 0.01$) was small (non-white cells in the lower triangle of Fig. 2), compared with 21.9 % (279/1275) and 27.19 % (31/114) of SSR markers being in linkage disequilibrium ($D' > 0$, $p < 0.05$) across the whole genome and within the same linkage groups, respectively.

The distance over which LD decays ($D' < 0.5$) affects the accuracy of association mapping and the number of markers required for such analysis. Using regression analysis, the D' value of pairwise syntenic markers and genetic distance (cM) was calculated using the equation: $y = a + b\ln(x)$, where $a = 0.5557$ and

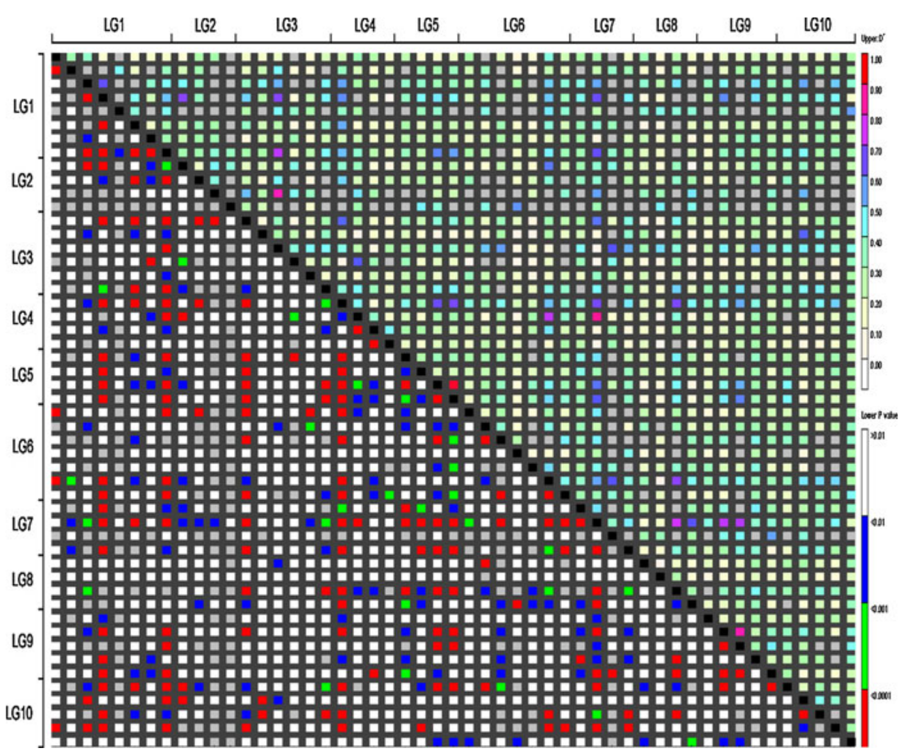

Fig. 2 Distribution of LD among 51 SSR loci across 10 sorghum chromosomes. SSR markers are organized in linkage groups (as in Table 2) marked along the X- and Y-axis; each pixel above the *diagonal* indicates the D' size of the corresponding marker pair as shown in the *color code* at the *upper right* while each pixel below the *diagonal* indicates the p value size of the testing LD of the corresponding marker pairs as shown in the *color code* at the *lower right*

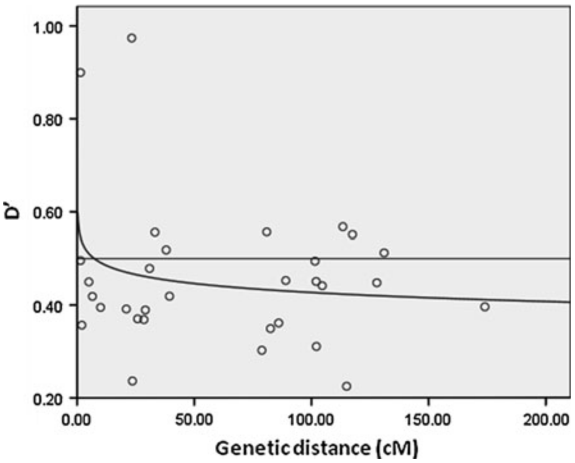

Fig. 3 Genome-wide LD (D′) distribution against genetic distance (cM). The simulation trend line showed that the D′ declined to below 0.5 within 10 cM

$b = -0.028$. The average genetic distance was about 7.31 cM when $D' = 0.5$ indicating that significant LD could extend 7.31 cM in this population (Fig. 3).

Phenotype-genotype association analysis

To reduce the influence of existing population structure, the individual Q value was treated as a covariate when the trait-marker association mapping was conducted. The three agronomic traits chosen for analysis—fresh stalk weight, stalk juice weight and brix—were continuously distributed in the populations and exhibited differences in mean between years and planting locations (Table 3). ANOVA showed that these differences were highly significant ($p < 0.01$) (Table 4).

The SSR marker, xtxp340, on chromosome 1 was detected to associate with sugar concentration (brix) in year 2010 at Shijiazhuang and Haixing, and explained 17.76 and 16.42 %, of total phenotypic variation for this trait at these two sites, respectively. This locus also tended to show an association with brix at Shijiazhuang in year 2011, although this association was not highly significant ($p = 0.0183$, >0.001). Three SSR markers, Undhsbm105, xtxp141 and xtxp273 on chromosome 10, 10 and 8, respectively, were detected to associate with fresh stalk weight (FSW). The xtxp141 marker was highly associated with FSW at Shijiazhuang station in both 2010 and 2011, explaining 35.95 and 29.50 % of the total phenotypic variance, respectively, while Undhsbml105 explained 29.17 % of total phenotypic variation at Shijiazhuang in year 2010. However, neither of these two markers were associated with FSW at Haixing station, where plants grew on saline/alkaline soils. In contrast, the xtxp273 marker on chromosome 8, was highly associated with FSW at this site, explaning 16.67 % of the trait variation. Markers Undhsbm105 and xtxp141 were also highly associated with stalk juice weight (SJW) at Shijiazhuang in both 2010 and 2011, explaining 25.62 and 28.21 % of phenotypic variation, respectively. However no marker loci were associated with the trait at Haixing Station in 2010 (Table 5).

Discussion

Sugar yield in sorghum is mainly determined by stalk sugar concentration and juice yield. Previous studies have found QTLs controlling brix and sucrose

Table 3 Phenotypic variation of fresh stalk weight (FSW, g), stalk juice weight (SJW, g) and brix in different years and locations

Trait	Years	Location	Mean	Max	Min	Std
FSW	2011	Shijiazhuang	315.09	1030.00	36.50	233.5
	2010	Shijiazhuang	321.47	1172.56	53.80	217.46
		Haixing	263.62	1048.00	23.33	198.99
SJW	2011	Shijiazhuang	103.80	428.50	1.75	107.82
	2010	Shijiazhuang	129.61	445.16	9.35	94.37
		Haixing	117.68	507.80	4.60	95.34
Brix	2011	Shijiazhuang	10.90	19.00	3.50	4.20
	2010	Shijiazhuang	12.59	21.42	3.40	4.27
		Haixing	11.05	19.67	1.65	5.48

Table 4 ANOVA of fresh stalk weight (FSW, g), stalk juice weight (SJW, g) and brix

Trait	Source of variation	df	MS	F value	p value
FSW	Year	1	1660.73	0.15	0.701
	Location	1	1266104.00	112.79	<0.0001**
	Accessions	117	380900.80	33.93	<0.0001**
	Accession × year	106	21645.62	1.93	<0.0001**
	Accession × location	104	71336.51	6.36	<0.0001**
	Error	1333	11225.67		
SJW	Year	1	135898.50	55.25	<0.0001**
	Location	1	72782.13	29.59	<0.0001**
	Accessions	117	76955.68	31.29	<0.0001**
	Accession × year	106	6964.79	2.83	<0.0001**
	Accession × location	104	15919.85	6.47	<0.0001**
	Error	1331	2459.84		
Brix	Year	1	438.01	98.49	<0.0001**
	Location	1	860.45	193.49	<0.0001**
	Accessions	117	222.67	50.07	<0.0001**
	Accession × year	106	25.55	5.75	<0.0001**
	Accession × location	104	38.08	8.56	<0.0001**
	Error	1332	4.45		

** Highly significant

Table 5 SSR markers associated with brix, SFW and SJW ($p < 0.001$)

Trait	Year	Location	Locus	LG	Position	p	R^2	Variation
Brix	2011	Shijiazhuang						
	2010	Shijiazhuang	xtxp340	1	219.6	1.96E-04	0.1776	0.1776
		Haixing	xtxp340	1	219.6	8.54E-04	0.1641	0.1642
FSW	2011	Shijiazhuang	xtxp141	10	129.1	2.20E-06	0.3142	0.3595
	2010	Shijiazhuang	Undhsbm105	10	22.2	8.89E-06	0.2820	0.2917
			xtxp141	10	129.1	1.93E-05	0.2949	0.295
		Haixing	xtxp273	8	0	7.34E-04	0.1667	0.1667
SJW	2011	Shijiazhuang	Undhsbm105	10	22.2	6.16E-04	0.2562	0.2562
			xtxp141	10	129.1	1.88E-06	0.3687	0.3687
	2010	Shijiazhuang	Undhsbm105	10	22.2	5.71E-06	0.2821	0.2821
		Haixing						

concentration on chromosomes 1, 2, 3, 4, 5 and 7, respectively, in different mapping populations. One of these QTLs, positioned close to a marker (Tx145) located on chromosome 1, was reported to explain 33.9 % of trait variation in a previous study (Lekgari 2010). In the present study, which involved an examination of 43 sweet sorghum and 76 grain sorghum accessions, an SSR marker located in the same region on chromosome 1 as TX145, was also identified to associate with brix, explaining 17.76 and 16.42 % of total trait variation, respectively, at two sites that differed markedly in soil conditions for plant cultivation.

Stalk juice yield and soluble solid content are two major components of total sugar yield (Corn 2009). Because percent soluble solids has a physiological limit of ~25 %, focusing on increased juice yield is the most efficient method of increasing total sugar production (Mangelsdorf 1958, in Felderhoff et al. 2012). Moisture content in stalks is thought to be directly associated with stalk types—pithy (dry) or juicy—which are genetically determined by one major gene, *d* (Hilson 1916, in Felderhoff et al. 2012). However, Felderhoff's study showed that the pithy trait is controlled by a major gene that determines the appearance of the midrib and a large portion of the visible stem structure, but does not influence juice yield as previously believed. In the present study, two SSR markers on chromosome 10 were found to associate with stalk juice weight (SJW) confirming that SJW was a quantitative trait. Felderhoff et al. (2012) showed that sorghum biomass yield, fresh stalk weight and stalk juice weight were significantly correlated and in this study we also found FSW and SJW to be highly correlated (data not shown). Interestingly, we showed that the xtxp141 marker on chromosome 10 was highly associated with FSW, while previous studies have mapped a biomass QTL to the same region (Lekgari 2010; Ritter et al. 2007).

In the present study different degrees of LD were detected among syntenic and nonsyntenic markers. This may be caused by historical recombination among different linkage groups or by pairwise epistatic interactions between distinct loci. By means of association analysis, four marker loci were detected to be associated with sugar yield-related traits, and some of these were consistent with previous QTL mapping results. Association mapping relies on ancestral recombination and natural genetic diversity within a population to analyze quantitative traits and is based on the LD concept. LD is affected by such biological factors as recombination and allelic frequencies, and historical factors that affect population size, such as selection, bottlenecks due to extreme genetic drift, and selection for or against a phenotype controlled by two non-linked loci (epistasis). Mating patterns and gene flow between individuals of genetically distinct populations followed by intermating can also strongly influence LD (Gómez et al. 2011). It is important to realize that the presence of high LD could lead to false associations and low mapping resolution. Thus characterizing patterns of linkage disequilibrium (LD) is critical for the design of association studies, interpretation of association peaks, and the transfer of alleles in marker-assisted selection (Morris et al. 2013). The average LD extension distance in the present sorghum population examined was estimated to be 7.31 cM, which is greater than that reported in other studies (Morris et al. 2013, Hamblin et al. 2005, Bouchet et al. 2012). This difference may be attributed to low genome coverage of markers and fewer genotypes surveyed in our study. Also, because sorghum is a predominantly selfing species, but readily outcrosses, greater extent of LD than in outcrossing species is expected.

Bouchet suggested that more than 100,000 markers may be required to perform genome-wide association studies in collections covering worldwide sorghum diversity after they analyzed 177 sorghum accessions with 1122 informative physically anchored DArT (diversity arrays technology) markers (Bouchet et al. 2012). Our use of 51 SSR markers has provided a preliminary understanding of sorghum genome linkage disequilibrium, average distance of LD decay and association with QTL affecting of sugar yield-related traits in a panel of accessions that included a large number of Chinese landraces. Many more markers will be needed for detailed genome wide association mapping analysis of such collections in the future.

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